

Skills, Inequality, and Referrals*

Experimental Design

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1 Background and Setting

Inequality and Social Class. Colombia has consistently ranked as one of the most unequal countries in Latin America (Barreto Herrera et al., 2024), with the richest decile earning 50 times more than the poorest decile (ECLAC, 2023). This economic disparity is reflected by a highly stratified society, where social classes remain separated by education, family background, lifestyle, occupation, power, and geographic residence.

Stratum system. In 1994, Colombia introduced a nationwide classification system dividing the population into 6 strata based on housing characteristics and neighborhood amenities. This system, initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), now extends to university fees and social program eligibility. While imperfect -because it disregards the income of residents, this system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008). We use this exogenous cutoff as the measure

*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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of social class in our experiment: Students in strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES.

Universidad Autónoma de Bucaramanga. Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia, with approximately 10,000 enrolled students. UNAB’s student body is notably diverse, with slightly more than half of the students classified as low-SES (see Appendix Figure A.1 for a detailed stratum distribution). This diversity reflects a broader trend in Colombian society, where even lower middle-class families (stratum 2 and 3) prioritize university education for their children despite the financial burden (Hudson & Library of Congress, 2010, p. 103). The university environment provides a unique opportunity for collaborative inter-class contact on equal status, whose positive effects on discrimination are casually documented in numerous studies (Lowe, 2021; Mousa, 2020; Rao, 2019). We draw our sample from a cohort of students taking two semester-long courses. Taking both courses are a requirement to graduate for all students at UNAB, ensuring diversity in academic programs and social classes. However, this approach may introduce some selection bias due to the sorting of students from different social classes into specific study programs.

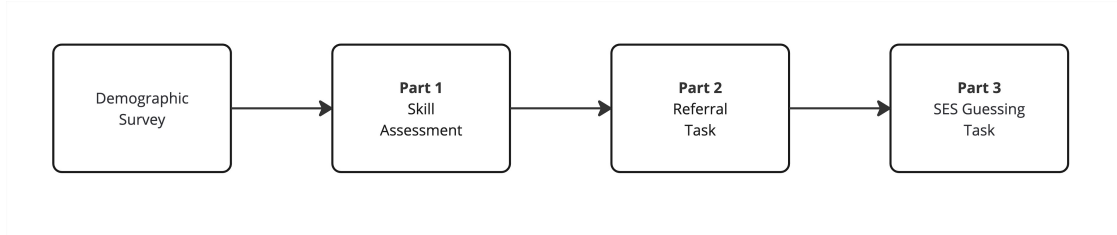
2 Design

We designed a referral experiment between students from different social classes, and experimentally varied the incentives to refer a low-SES peer. The study design consists of a single experiment with three sequential parts with sessions organized at the classroom level (see Figure 1). The instructions are provided in Appendix B.

2.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable

Figure 1: Experiment Timeline



Note: Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

in a context with university students.¹ This approach ensures that we observe the distribution of skills in our sample, ask for referrals on these skills, and compare how well the same individuals refer peers for a specific skill. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.²

Cognitive Skills. We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the first 9 items and 8 thereafter.

¹For the relationship between wages and social skills conditional on cognitive skills, see Deming (2017); Edin, Fredriksson, Nybom, and Öckert (2022); Kuhn and Weinberger (2005); Weinberger (2014).

²The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

Social Skills. We measure social skills with the Multiracial Reading the Mind in the Eyes Test (MRMET) from Kim et al. (2022).³ The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

2.2 Referral Task

We create a referral task to measure bias against low-SES peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following page, we provide an alphabetically ordered list of all classmates. Using this list instead of a free text field could help alleviate biases related to recall in referral choice *cite who?*. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.⁴

Treatment. We have one between subject treatment, and it varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on

³We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

⁴We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

performance. This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure A.4 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

2.3 SES Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure A.5 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish SES independent of test performance, as SES identification is relevant to our study.

3 Sample, Incentives, and Procedure

Sample. We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 713 individuals who completed the study protocol, resulting in an 83% participation rate. Table 1 presents key demographic characteristics and academic performance indicators across treatments (Appendix Table A.1 compares the sample and missing students). The sample is well-balanced between the Baseline and Quota conditions. We observe no statistically significant differences in any of the reported variables (all p values > 0.1), indicating balanced randomization. The sample is characterized by a majority of lower-SEB students. About one-third of the sample are first-generation college students. The gender distribution is balanced. The mean GPA of 3.95 is consistent across both treatments.

Incentives. Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill

tests. 20% of participants were eligible for the bonus. We randomly picked one skill test for each eligible participant and drew a number between 1 and 100. The participant received the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% ($=1/18$) for the Cognitive Skill test and by 2.78% ($=1/36$) for the Social Skill test. In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate.

Procedure. Data collection occurred during the last weeks of April 2024. Our local partner, the Social Bee Lab, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

Table 1: Comparison of demographic characteristics and academic outcomes across treatment conditions

	Baseline	Quota	<i>p</i>
Low-SES	59%	55%	0.297
Cognitive skill (out of 18)	10.04	10.27	0.322
Social skill (out of 36)	18.45	18.50	0.886
GPA	3.95	3.95	0.828
Entry Exam	61.85	62.17	0.638
Age	19.33	19.02	0.228
Female	52%	47%	0.195
First Generation	34%	37%	0.386
Ethnic Minority	1%	3%	0.133
Rural Community	30%	27%	0.308
Has Scholarship	1%	1%	0.916
# Semesters at UNAB	3.18	3.17	0.916

Note: Low-SES, Female, First Generation, Ethnic Minority, Rural, and Has Scholarship represent binary outcomes and *p*-values for these variables are from two-sample tests of proportions. For continuous variables (Age, GPA, Cognitive skill, Social skill, Entry Exam, and Semester), *p*-values are from two-sample t-tests with equal variances. All reported *p*-values are two-tailed. Sample sizes vary slightly across measures due to missing data.

4 Empirical Analysis

4.1 Individual Performance

Our first goal is understanding which individuals get referred more often. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.⁵ We consider the fraction of referrals received by total number of referrals that could have been received instead (see Appendix Figure A.2 for the distribution). This approach combines the intensive and extensive margins and also makes comparisons across classrooms with different sizes easier.⁶ Our three dependent variables are fractions of: (i) cognitive skill referrals, (ii) social skill referrals, and (iii) all referrals.

Under performance pay, higher performing classmates in skill tests should collect more referrals. However, it is not clear whether skills can be observed by referrers.⁷ We therefore include other common academic performance measures that may affect our dependent variables. These measures are standardized scores of GPA and centralized entry exam to the university, similar to SAT/ACT. Regressions include $\mathbf{X}_i$ for all the remaining individual correlates from Table 1. We estimate three referral fractions y_i :

$$y_i = \beta_0 + \beta_1 GPA_i + \beta_2 E.Exam_i + \beta_3 Cognitive_i + \beta_4 Social_i + \mathbf{X}_i\gamma + \epsilon_i \quad (1)$$

Table 2 illustrates our first set of findings. Our comparison of interest is between skills and other academic performance measures. Conditional on academic performance and individual correlates specified above, there is no statistically significant difference between skill level and the fraction of referrals received for that skill (columns 1-2). This holds true also when both skills are aggregated (column 3). We emphasize aggregated fraction of referrals because most referrals overlap across skills, illustrated in Appendix

⁵Only 86 of the 849 students (10%) never get a referral for either skill.

⁶The number of participating students in a classroom mechanically drives the number of total referrals that could be received by a classmate.

⁷We did not provide skill feedback during the experiment. Feedback during sessions could have contaminated referral choices because participants would have incentive to share skill test results.

Figure A.3. Point estimates for skills are very close to zero. The 95% confidence intervals rule out that a one standard deviation increase in either skill causes more than a 1 percentage point difference in the fraction of referrals received. This is an order of magnitude smaller margin when compared to the effects of a one standard deviation increase in GPA or Entry Exam.

A one standard deviation increase in GPA increases the fraction of referrals by about 3 percentage points on a base rate of 11% across the three dependent variables. This is a modest 27-percent increase in the fraction of referrals received.⁸ A one standard deviation increase in the Entry Exam score increases the fraction of referrals by about 2.4 percentage points for cognitive skills, with a markedly lower 1.1 percentage points for social skills. Smaller point estimates on performance metrics and lower explained variance suggests our referral model performs better for cognitive skill. Social skill referrals are seemingly made with other considerations not entirely captured in the model. The results suggest participants are not referring classmates who score high on skill tests but refer instead based on their academic performance.⁹ This may be due to classmates being able to observe academic performance of peers but not their skill test results.

4.2 Social Distance

Our second goal is understanding which individuals get referred more often, conditional on performance and social distance. As a measure of social distance between classmates, we consider the standardized fraction of classmates from the same academic program.¹⁰ To address concerns about selection into courses by program we include classroom fixed effects.¹¹ Our dependent variable is the fraction of referrals received by total number

⁸The average participant in the average classroom with about 21 students receives 2.3 referrals. A one standard deviation increase in GPA increases this number to 2.9 referrals.

⁹Our findings are robust to: (i) restricting the observations to the baseline condition (see Appendix Table A.2), and (ii) including classroom fixed effects (Appendix Table A.3).

¹⁰There is a large overlap between faculties and programs, and for granularity we prefer the latter.

¹¹Students choose when to take the two sampled mandatory courses depending on how it fits into their program-specific schedules. Since different programs have different course schedules, students select into specific classes that best accommodate their timetables. Without classroom fixed effects, our estimates

Table 2: Fraction of referrals conditional on performance and social distance

	(1)	(2)	(3)
Dep. Var.	Cognitive	Social	All
GPA	0.03253*** (0.00471)	0.02759*** (0.00494)	0.03006*** (0.00444)
Entry Exam	0.02455*** (0.00596)	0.01113* (0.00583)	0.01784*** (0.00558)
Cognitive	-0.00413 (0.00535)	-0.00412 (0.00517)	-0.00413 (0.00499)
Social	0.00267 (0.00414)	0.00219 (0.00434)	0.00243 (0.00386)
Constant	0.11979*** (0.04292)	0.10607*** (0.03959)	0.11293*** (0.03831)
Individual Controls	Yes	Yes	Yes
Observations	631	631	631
R-squared	0.162	0.107	0.147

Note: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the fraction of referrals received relative to potential referrals for cognitive (1), social (2), and all referrals (3). Controls include first generation status, SES, gender, ethnicity, rural status, age, and semester. Sample is restricted to participants with complete administrative and experimental data.

of referrals that could have been received averaged across skills. Remaining individual correlates are included in $\mathbf{X}_i$. We estimate referral fraction y_i :

$$y_i = \beta_0 + \beta_1 GPA_i + \beta_2 E.Exam_i + \beta_3 Program_i + \mathbf{X}_i\gamma + \epsilon_i \quad (2)$$

While not our main focus, one standard deviation increase in the fraction of classmates from the same academic program increases the fraction of referrals by about 1.1 percentage points across the three dependent variables (10% increase). This effect size is comparable to the impact of the academic performance measures and considerably higher than skill test scores.

4.3 Social Class

In our second empirical specification, we document whether there is social class homophily in referrals, conditional on the relevant performance measures identified earlier.¹² Specifically, we create a matched dataset between referrers and referrals for skills. We estimate social class y_{ij} for referral i from referrer j :

$$y_{ij} = \alpha + \beta_1 GPA_j + \beta_2 E.Exam_j + \beta_3 ReferrerSES_i + \beta_4 ShareSES_i + \beta_5 Program_i + \mathbf{X}_i\gamma + \epsilon_i \quad (3)$$

Our binary dependent variable takes value 1 if a referral was to a low-SES classmate and 0 otherwise. From our first specification, we include as independent variables referral GPA and entry exam scores for performance measures. Referrer SES is a dummy variable taking value 1 if the referrer is low-SES and 0 otherwise. As a measure of the SES heterogeneity in referrer choice sets, we include the fraction of low-SES classmates

Hypothesis 1 - Social class bias

H1a. Does the probability of being referred depend on the social class of the potential for social distance and referral decisions suffers from omitted variable bias.

¹²See [McPherson, Smith-Lovin, and Cook \(2001\)](#) for a discussion on homophily in general, for evidence on social class homophily see [Chetty et al. \(2022a, 2022b\)](#). For experimental evidence on a different type of homophily (gender) in referrals see [Beaman, Keleher, and Magruder \(2018\)](#).

nominees after controlling for their performance and the social class composition at the session level? In other words, is there social class bias in referrals?

H1b. Which social group makes more referrals to their own social class, controlling for performance and the social class composition at the session level? Do the lower or higher social class individuals drive the bias?

5 hypotheses-to be removed

5.1 Main Analysis: Social class and referral choices across conditions

We then examine whether the quota condition causes changes in referral compositions regarding social classes. We are also interested in understanding the equity-efficiency tradeoff of quotas, if there is one. Specifically:

Hypothesis 2 - Social class quota

H2a. Does the quota cause an increase in the number of lower social class referrals, controlling for the social class composition at the session level?

H2b. Does the quota cause a decrease in referral quality, controlling for the social class composition at the session level?

H2c. Does the impact of the quota on referral quality and the number of lower social class referrals depend on whether the referrals are for cognitive or social skills?

5.2 Main Analysis: Mechanisms driving bias

We will start by looking at referral performance. Experimental evidence has shown that referrals from high-performing individuals perform better in tasks than others ([Beaman et al., 2018](#); [Beaman & Magruder, 2012](#); [Pallais & Sands, 2016](#)). More precisely:

Hypothesis 3 - Referrer Performance

H3a. Do referrers with higher cognitive and/or social skills make higher quality referrals?

H3b. Do referrers with higher cognitive and/or social skills have a smaller social class bias when making referrals?

5.3 Exploratory Analysis

We examine the remaining variables to complement the findings for our primary research questions. What other types of homophily do we observe in our sample? How does the referrer-referral performance correlation depend on other characteristics?

Hypothesis 4 - Additional Variables

H4a. Does the probability of being referred depend on other characteristics of the potential nominees (e.g., age, GPA, national university entry exam score, parental education, gender, co-enrollment history, ...), controlling for performance and these characteristics at the session level?

Hba. Does the quality of referrals depend on other characteristics of the referrer (e.g., age, parental education, GPA, time spent at the university,...)?

Evidence suggests social skills are harder to observe for companies (Bassi & Nansamba, 2022). If referrers are good at discerning social skills, this could make resorting hiring via referrals a tempting choice for companies. Referrals from high-performing individuals perform better in the same task when provided with feedback (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016). We did not provide such performance feedback to participants. Nevertheless, we had priors about how well contact would make students predict the performance of their classmates for each skill. We assumed that Cognitive skills would be easier to detect than Social skills. With this in mind, we

References

Baron-Cohen, S., Wheelwright, S., Hill, J., Raste, Y., & Plumb, I. (2001). The “Reading the Mind in the Eyes” Test Revised Version: A Study with Normal Adults, and

- Adults with Asperger Syndrome or High-functioning Autism. *The Journal of Child Psychology and Psychiatry and Allied Disciplines*, 42(2), 241–251. (Publisher: Cambridge University Press) doi: 10.1017/S0021963001006643
- Barreto Herrera, K. Y., Castellanos Rodriguez, L. E., Garcia Garcia, C., Maquera Sardon, D. A., Montoya Munoz, K. Y., Nopo Aguilar, H. R., ... Winkler-Seales, H. J. (2024). *Regional poverty and inequality update spring 2024* (Poverty and Equity Global Practice Brief). Washington, D.C.: World Bank Group. Retrieved from <http://documents.worldbank.org/curated/en/099070124163525013/P17951815642cf06e1aec4155e4d8868269>
- Bassi, V., & Nansamba, A. (2022). Screening and signalling non-cognitive skills: experimental evidence from uganda. *The Economic Journal*, 132(642), 471–511.
- Beaman, L., Keleher, N., & Magruder, J. (2018). Do Job Networks Disadvantage Women? Evidence from a Recruitment Experiment in Malawi. *Journal of Labor Economics*, 36(1), 121–157. doi: 10.1086/693869
- Beaman, L., & Magruder, J. (2012). Who Gets the Job Referral? Evidence from a Social Networks Experiment. *American Economic Review*, 102(7), 3574–3593. doi: 10.1257/aer.102.7.3574
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebe, J., Hendren, N., Fluegge, R. B., ... others (2022a). Social capital ii: determinants of economic connectedness. *Nature*, 608(7921), 122–134.
- Chetty, R., Jackson, M. O., Kuchler, T., Stroebe, J., Hendren, N., Fluegge, R. B., ... others (2022b). Social capital i: measurement and associations with economic mobility. *Nature*, 608(7921), 108–121.
- Deming, D. J. (2017). The growing importance of social skills in the labor market. *The quarterly journal of economics*, 132(4), 1593–1640.
- ECLAC. (2023). *Social panorama of latin america and the caribbean 2023: labour inclusion as a key axis of inclusive social development*. United Nations. Retrieved from <https://www.cepal.org/es/publicaciones/68702-panorama-social-america-latina-caribe-2023-la-inclusion-laboral-como-eje-central>

- Edin, P.-A., Fredriksson, P., Nybom, M., & Öckert, B. (2022). The rising return to noncognitive skill. *American Economic Journal: Applied Economics*, 14(2), 78–100.
- Guevara S, J. D., & Shields, R. (2019). Spatializing stratification: Bogotá. *Ardeth. A Magazine on the Power of the Project*(4), 223–236.
- Hudson, R. A., & Library of Congress (Eds.). (2010). *Colombia: a country study* (5th ed.). Washington, D.C: Federal Research Division, Library of Congress: For sale by the Supt. of Docs., U.S. G.P.O. Retrieved from the Library of Congress, <https://www.loc.gov/item/2010009203/>.
- Kim, H., Kaduthodil, J., Strong, R. W., Germine, L., Cohan, S., & Wilmer, J. B. (2022). *Multiracial Reading the Mind in the Eyes Test (MRMET): an inclusive version of an influential measure* (preprint). doi: 10.31219/osf.io/y8djm
- Kuhn, P., & Weinberger, C. (2005). Leadership skills and wages. *Journal of Labor Economics*, 23(3), 395–436.
- Lowe, M. (2021). Types of contact: A field experiment on collaborative and adversarial caste integration. *American Economic Review*, 111(6), 1807–1844.
- McPherson, M., Smith-Lovin, L., & Cook, J. M. (2001). Birds of a feather: Homophily in social networks. *Annual review of sociology*, 27(1), 415–444.
- Mousa, S. (2020). Building social cohesion between christians and muslims through soccer in post-isis iraq. *Science*, 369(6505), 866–870.
- Pallais, A., & Sands, E. G. (2016). Why the Referential Treatment? Evidence from Field Experiments on Referrals. *Journal of Political Economy*, 124(6), 1793–1828. doi: 10.1086/688850
- Rao, G. (2019). Familiarity does not breed contempt: Generosity, discrimination, and diversity in delhi schools. *American Economic Review*, 109(3), 774–809.
- Raven, J. C. (1936). *The Performances of Related Individuals in Tests Mainly Educative and Mainly Reproductive Mental Tests Used in Genetic Studies* (PhD Thesis). University of London (King’s College).
- Raven, J. C., Raven, J., & Court, J. H. (1976). *Standard progressive matrices: sets A*,

- B, C, D & E.* Oxford, UK: Oxford Psychologists Press.
- Schilbach, F., Schofield, H., & Mullanathan, S. (2016). The Psychological Lives of the Poor. *American Economic Review*, 106(5), 435–440. doi: 10.1257/aer.p20161101
- Uribe-Mallarino, C. (2008). Estratificación social en bogotá: de la política pública a la dinámica de la segregación social. *Universitas humanistica*(65), 139–172.
- van Leeuwen, B., Noussair, C. N., Offerman, T., Suetens, S., van Veelen, M., & van de Ven, J. (2018). Predictably Angry—Facial Cues Provide a Credible Signal of Destructive Behavior. *Management Science*, 64(7), 3352–3364. doi: 10.1287/mnsc.2017.2727
- Weidmann, B., & Deming, D. J. (2021). Team Players: How Social Skills Improve Team Performance. *Econometrica*, 89(6), 2637–2657. doi: 10.3982/ECTA18461
- Weinberger, C. J. (2014). The increasing complementarity between cognitive and social skills. *Review of Economics and Statistics*, 96(5), 849–861.

A Additional Figures and Tables

A.1 Additional Figures

Figure A.1: Stratum distribution of the sample

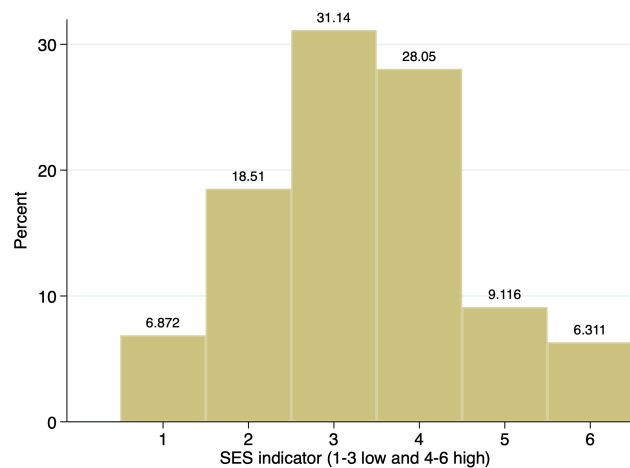
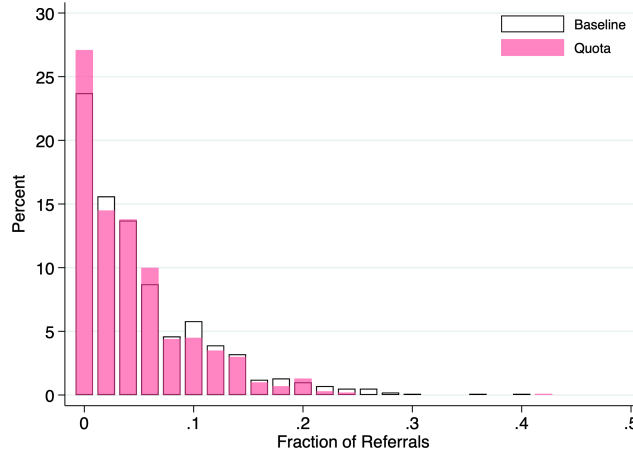
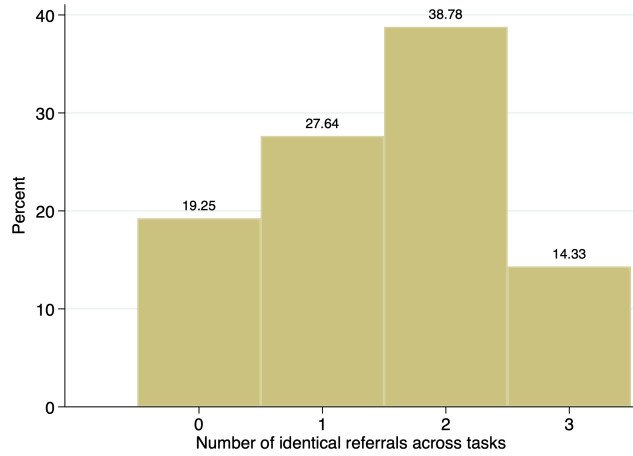


Figure A.2: Fraction of referrals received



Note: Fraction of referrals received for both skills by the total number of referrals that could have been received. Figure disaggregated across baseline and quota treatments.

Figure A.3: Distribution of overlapping referrals for skills



Note: The first bar (value of 0) indicates the percentage of individuals with all unique referrals. The last bar (value of 3) indicates the percentage of individuals with identical referral choices for both skills. Data excludes referrals that were made to one's own. Most individuals have 2 overlapping referral choices.

A.2 Additional Tables

Table A.1: Comparison of Variables Between the Sample and Missing Students

	Sample	Missing	p
Ratio referrals (both skills)	0.127	0.043	0.000
GPA (standardized)	0.044	-0.273	0.001
Entry Exam (standardized)	0.028	-0.168	0.046
# Semesters at UNAB	3.171	3.188	0.884
Age	19.182	20.287	0.001
Female	49.8%	48.5%	0.788
Ethnic Minority	2.1%	4.4%	0.114
Rural Community	28.8%	31.6%	0.501
Has Scholarship	0.8%	0.7%	0.899

Note: Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. p -values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, p -values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported p -values are two-tailed.

Table A.2: Fraction of referrals in the baseline condition conditional on performance

	(1)	(2)	(3)
Dep. Var.	Cognitive	Social	All
GPA	0.0202*** (0.0032)	0.0180*** (0.0030)	0.0191*** (0.0028)
Entry Exam	0.0119*** (0.0034)	0.0056 (0.0038)	0.0088*** (0.0034)
Cognitive	0.0003 (0.0031)	-0.0024 (0.0031)	-0.0010 (0.0029)
Social	0.0001 (0.0025)	-0.0008 (0.0027)	-0.0004 (0.0023)
Constant	0.0672*** (0.0255)	0.0344 (0.0239)	0.0508** (0.0221)
Controls	Yes	Yes	Yes
Observations	631	631	631
R-squared	0.151	0.096	0.138

Note: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables are the fraction of referrals received relative to potential referrals for cognitive (1), social (2), and all referrals (3). Controls include first generation status, SES, gender, ethnicity, rural status, age, and semester. Sample is restricted to participants with complete administrative and experimental data.

Table A.3: Fraction of referrals conditional on performance with classroom fixed effects

	(1)	(2)	(3)
Dep. Var.	Cognitive	Social	All
GPA	0.0336*** (0.0049)	0.0281*** (0.0054)	0.0305*** (0.0046)
Entry Exam	0.0284*** (0.0058)	0.0136** (0.0057)	0.0200*** (0.0053)
Cognitive	-0.0043 (0.0050)	-0.0050 (0.0048)	-0.0051 (0.0045)
Social	-0.0014 (0.0042)	-0.0006 (0.0044)	-0.0001 (0.0039)
Constant	0.1309*** (0.0400)	0.1148*** (0.0387)	0.0977*** (0.0351)
Classroom FE	Yes	Yes	Yes
Individual Controls	Yes	Yes	Yes
Observations	631	631	631
R-squared	0.290	0.218	0.301

Note: Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Dependent variables for columns 1 and 2 are fractions of referrals received out of all potential referrals for cognitive and social skills. Dependent variable for column 3 is the average of both skill referral fractions. Sample is restricted to participants with complete administrative and experimental data.

B Experiment

We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.

Please enter the password:

[classroom-specific password sent to each participant the day before data collection]

Welcome

Welcome to this study organized by the Social Bee Lab. You have been invited to participate in a survey where you can make a series of decisions. The study takes approximately 20 minutes to complete. During the study, you should not communicate with any other students. If you have any questions at any time, please raise your hand. One of the assistants will help you privately.

In this study, you can win bonus money depending on your choices. In total, we will draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos among the participants of this classroom. It is also possible for the same person to win more than one voucher. The following screens will detail how the bonus draw will be conducted. The UNAB finance office will make the payment of the vouchers through Nequi.

All your decisions in this survey will be anonymized. Therefore, the answers you provide will not affect your grades in this class or your records at the university. We will use your personal information to determine the bonus allocation, but after that, we will remove

any data that identifies you.

This survey has several parts. Each of these parts has specific instructions. Please read the instructions for each part carefully because they describe how you can earn bonuses. This study has been approved by the [omitted for anonymous review] on the condition that all the information we provide is true and all the bonuses we offer are real.

On the next screen, we present you with an informed consent form that you must accept to participate in this study.

Informed Consent

You have been invited to participate in a study to learn more about how people make decisions in common scenarios.

This study is conducted by [omitted for anonymous review] and the Social Bee Lab at UNAB. The purpose of this study is to broaden our understanding of how people make decisions.

Participation in this study is voluntary. You may opt-out at any time. No known risks are associated with your participation in this project beyond those of everyday life. Apart from the monetary bonuses that will be drawn, participation has no direct benefits.

The Social Bee Lab is in charge of data collection. Your answers in this study are anonymous and will not be shared with anyone. In addition to your answers, UNAB will provide the Social Bee Lab with administrative records of your courses and your university entrance exam score. Your records, decisions, and your identity will be kept strictly confidential. Data about you collected within the scope of the study are used for scientific purposes only and are treated as strictly confidential. The Social Bee Lab will

anonymize your data, and the researcher will analyze it without knowing your identity. All data generated will be stored on the researcher's computer. You have the right to access your personal data and request its deletion. You can exercise this right by contacting the researcher.

If something is unclear or you have any questions, you can contact [omitted for anonymous review].

If you have questions about your rights as a participant, you can contact [omitted for anonymous review].

By continuing to the next screen, you agree to participate in this study.

Before you start, please answer these four questions.

What is your gender?

[Male, Female]

What is the socio-economic stratum to which your family belongs?

[Stratum 1 to Stratum 6]

What is your father's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

What is your mother's highest acquired level of education?

[Primary school, High school, Technical school, Undergraduate, Graduate, Postgraduate, Not applicable.]

Part 1

You will now participate in two quizzes, each lasting five minutes. Please try to answer them to the best of your ability.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained below.

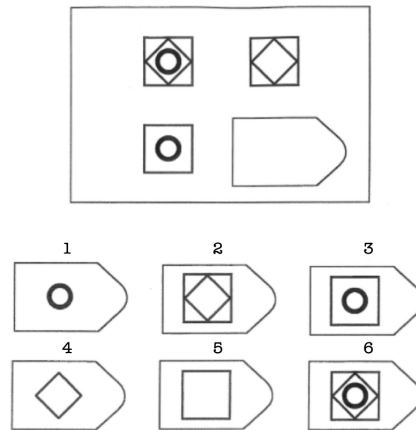
[classroom-specific illustrations explaining the incentive structure]

[random assignment to either cognitive or social skills test]

Test - Cognitive Skill

In this test, you will see a series of images. Below is an example of the images you will solve. At the top of each image, there is a pattern with a piece that has been removed.

Your task is to choose which of the six pieces completes the pattern correctly. For each image, there is only one correct piece. Look at the following example:



First, notice a square in the upper left, the upper right, and the lower left. Also, notice that the circle is eliminated when one moves from the upper left to the upper right. Finally, the rhombus is eliminated when moving from the upper left to the lower left. Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a square. So, the correct answer is piece 5.

To give your answer to each image, you must choose the correct option and then continue to the next screen. After giving your answer you cannot go back.

You will have 5 minutes to complete the test, which consists of 18 images to solve. The percentage of correct answers will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

Problem 1

[screenshot of Raven's matrix]

[After participants submit an answer, a new matrix appears on the screen. The sequence of matrices is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 18 matrices.]

You have finished the test. You can proceed to the next screen.

How did you do on the test?

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct problems than you?

[Slider from 0 to 10]

Test - Emotions

In this test, you will see a series of photographs. Below is an example of the pictures you will see. In each picture, you will see the eyes of a person. Below the picture, you will see four possible emotions that this person is feeling. Your task is to choose which of the four emotions correctly describes what the person is feeling. For each picture, there is only one emotion. Look at the following example:



[Happy, Disappointed, Shocked, Worried]

In this case, the correct answer is: Shocked.

To give your answer to each picture, you must choose the correct option and then continue to the next screen. After giving your answer you will not be able to go back.

You will have 5 minutes to complete the test, which consists of 36 photographs to solve. The percentage of correct answers you get will determine your chances of winning one of the 100.000 pesos bonuses if you are chosen for the drawing.

Are you ready?

Your 5 minutes will start as soon as you move to the next screen.

Photograph 1: Choose the word that best describes the photograph

[photo from Multiracial Reading the Mind in the Eyes Test]

[After participants submit an answer, a new photo appears on the screen. The sequence of photos is the same for all participants. Participants cannot return to a previous screen. Participants do not have to provide answers for all 36 photos.]

You have finished the test. You can proceed to the next screen.

How did you do on the test?

If we randomly choose 10 participants from this classroom, how many people do you think solved fewer correct photographs than you?

[Slider from 0 to 10]

Part 2

At the beginning of this study, all participants took two tests, one on cognitive ability and one on emotions. In this part, we will ask you to recommend the people who in your opinion will score the best on each test.

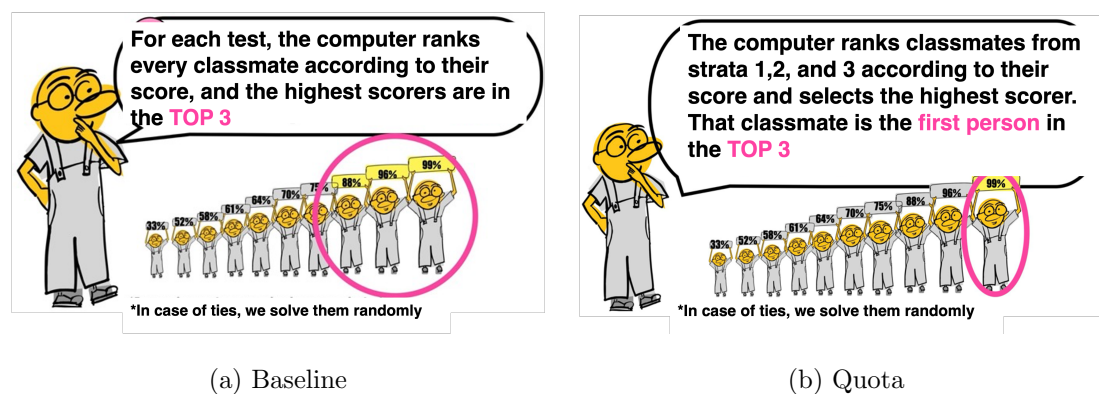
You may recommend 3 people per test, but you may not recommend yourself.

We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos for Part 2. The steps for allocating bonuses are explained below.

[random assignment to either quota or baseline condition]

[classroom-specific illustrations explaining the incentive structure depending on assignment to either baseline or quota conditions]

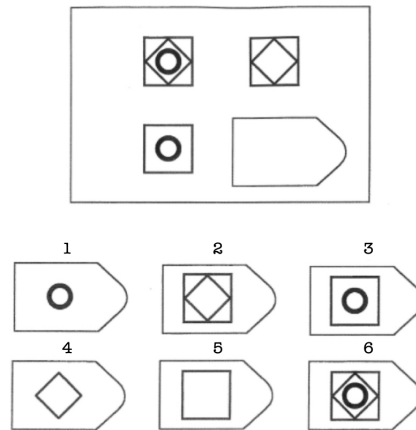
Figure A.4: Illustrations for the two conditions



[random assignment to either cognitive or social skills referral task]

Recommendation - Cognitive Skill

All participants took a test to identify the missing pattern in each image, as in the example below. This test is used to measure general intelligence.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend the three people you think will score the highest on the general intelligence test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

Select the students in this classroom who you consider to have the highest scores on the general intelligence test. (Select 3 students)

[Classroom-specific list of all classmate names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

Recommendation - Emotions

All participants took a test where they had to identify the emotion that best described the expression of each image as in the example below. This test is used to measure social skills.



Next, we will present you with a list of the names of all the students in this room. We will ask you to recommend 3 people you think will score the highest on the social skills test.

If you are chosen by the computer, each of your recommendations in the top 3 increases your chances of winning one of the 100.000 pesos bonuses.

Select the students in this classroom who you consider to have the highest scores on the social skills test. (Select 3 students)

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

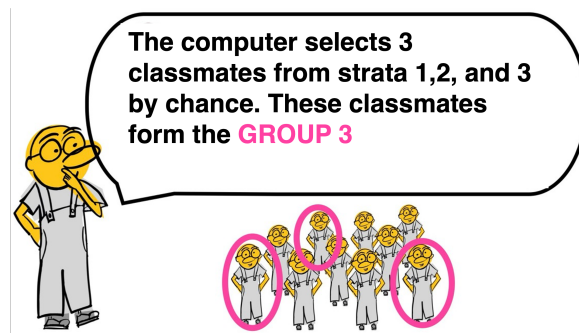
Part 3: Recommendation - Random draw

In this part, the computer will randomly choose three students who belong to strata 1, 2, or 3. We will ask you to nominate three people you think the computer will choose.

We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below.

[classroom-specific illustrations explaining the incentive structure]

Figure A.5: Illustration for the Guessing Task



Select the students in this classroom who belong to strata 1, 2, or 3, who you think will be randomly selected by the computer (Select 3 students).

[Classroom-specific list of all the names visible on one screen. Participants have to pick 3 classmates to continue. Picking their own name invalidates their choices.]

Part 4

Do you want to know your scores on the general intelligence test and the social skills test? We can analyze the data and give you a report that explains your strengths in these two areas. Also, what do these strengths mean, and how can you leverage them for your personal and professional development?

If you want to receive your skills report, we need to contact you again. We also want to be able to invite you to new studies where you can participate for more bonus money. Please indicate if you agree to be contacted again.

[I can be contacted for new studies and to send me my report. I can be contacted to send my report, but not for new studies. No, I do not want to be contacted again.]

[if participant gives consent to be contacted again]

Please enter your contact email:

[student email]